FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Advanced)-2019

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01 – 06, 19 – 24, 37 – 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 – 46) *contains 12 questions. Each question has* ***TWO (02)*** *matching lists:* ***LIST-I*** *and* ***LIST-II****.* ***FOUR*** *options are given representing matching of elements from* ***LIST-I*** *and* ***LIST-II****.* ***ONLY ONE*** *of these four options corresponds to a correct matching.*
   For each question, choose the option corresponding to the correct matching.
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

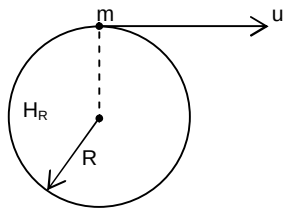
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A small particle is thrown parallel to the earth's surface with initial velocity μ wrt earth assuming there is no other planet and that the velocity of the earth is zero, then which of the following will be correct



(A) For the particle to move in circular orbit around the earth minimum value of 'μ' must be $\sqrt{\frac{GM}{R}}$

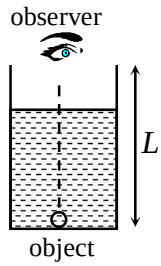
(B) For the particle to escape, minimum μ should be equal to $\sqrt{\frac{2GM}{R}}$

(C) For particle to escape, minimum μ should be more than $\sqrt{\frac{2GM}{R}}$

(D) The total mechanical energy of the particle and the earth system remain conserved

2. Let I_A and I_B be moments of inertia of a body about two axes *A* and *B* respectively. The axis *A* passes through the centre of mass of the body but *B* does not, then
(A) I_A may be greater than I_B
(B) I_B may be greater than I_A
(C) I_A may be equal to I_B
(D) I_A must be smaller than I_B

3. A cylinder is filled with a liquid of refractive index μ. The radius of the cylinder is decreasing at a constant rate *K*. The volume of the liquid inside the container remains constant at *V*. The observer and the object *O* are in a state of rest and at a distance *L* from each other. The apparent velocity of the object as seen by the observer, (when radius of cylinder is *r*)



(A) $\frac{(1-\mu)2KV}{(\pi\mu r^3)}$ (B) $\frac{(1-\mu)2KV}{(\pi\mu Lr^2)}$

(C) $\frac{(1-\mu)2K}{\mu}$ (D) $\frac{(1-\mu)K}{2\mu}$

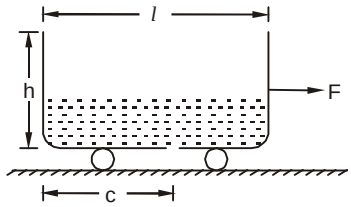
Space for Rough work

4. Nucleus A decays to B with decay constant λ_1 and B decays to C with decay constant λ_2. Initially at $t = 0$, number of nuclei of A and B are $2N_0$ and N_0 respectively. At $t = t_0$, number of nuclei of B stop changing. If at this instant number of nuclei of B are $\frac{3N_0}{2}$.

(A) the value of t_0 is $\frac{1}{\lambda_1}\ln\frac{4}{3}\frac{\lambda_1}{\lambda_2}$ (B) the value of t_0 is $\frac{1}{\lambda_2}\ln\frac{4}{3}\frac{\lambda_1}{\lambda_2}$

(C) the value of N_A at t_0 is $\frac{3N_0}{2}\frac{\lambda_2}{\lambda_1}$ (D) the value of N_A at t_0 is $\frac{2N_0}{3}\frac{\lambda_2}{\lambda_1}$

5. A rectangular vessel of dimension ($l \times b \times h$) and mass M contains a liquid of density ρ. The vessel has a orifice at its bottom at a distance c from the rear wall as shown in fig.



(A) The maximum volume of the water that can be stored when the vessel is accelerated is $\frac{hcb}{2}$.

(B) The maximum volume of the water that can be stored the when the vessel is accelerated is $\frac{hlb}{2}$

(C) Force F that must be applied when maximum water is stored is $\left[M+\frac{hcb\rho}{2}\right]\frac{hg}{c}$

(D) Force F that must be applied when maximum water is stored is $\left[M+\frac{hcb\rho}{2}\right]\frac{lg}{c}$

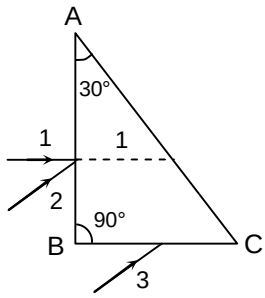
6. A closed vessel contains a mixture of two diatomic gases A and B. Molar mass of A is 16 times that of B and mass of gas A contained in the vessel is 2 times that of B. Which of the following statements are correct?
(A) Average kinetic energy per molecule of A is equal to that of B
(B) Root-mean-square value of translational velocity of B is four times that of A
(C) Pressure exerted by B is eight times of that exerted by A
(D) Number of molecules of B, in the cylinder, is eight times that of A

Space for Rough work

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST–I** and **LIST–II**. **FOUR** options are given representing matching of elements from **LIST–I** and **LIST–II**. **ONLY ONE** of these four options corresponds to a correct matching.

7. ABC is a right-angled prism kept in air. A ray (1) is incident on the face AB along the normal. Refractive index of the material of prism is the minimum value that will be required so that ray (1) undergoes total internal reflection at the face AC. Another ray (2) is incident on the face AB such that it emerges from face AC along the normal to AC. A third ray (3) falls on the face BC and emerges from face AC such that its angle of emergence is the same as that of incidence. Assuming light (1), (2) and (3) have the same wavelength, then match the following.



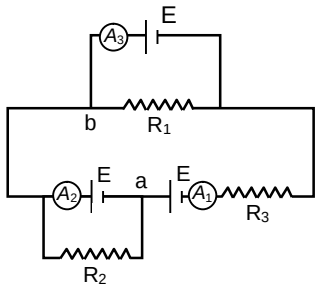
List–I		List–II	
(P)	Refractive index of the material of prism is	(1)	120
(Q)	Angle of incidence in degree of ray (2) is	(2)	90
(R)	Deviation in degree suffered by ray (2) is	(3)	2
(S)	Deviation in degree suffered by ray (3) is	(4)	60
		(5)	1.5

Codes:

(A) P–3, Q–2, R–1, S–2 (B) P–3, Q–2, R–4, S–1

(C) P–4, Q–1, R–2, S–3 (D) P–3, Q–1, R–4, S–2

8. In the circuit shown in figure, $R_1 = R_2 = R_3 = 3\Omega$ and e.m.f. of each cell is E = 4V and negligible internal resistance. All ammeters are ideal. Match the following:



List–I		List–II	
(P)	Reading of ammeter A_1 in ampere is	(1)	4/3
(Q)	Reading of ammeter A_2 in ampere is	(2)	8/3
(R)	Reading of ammeter A_3 in ampere is	(3)	4
(S)	Potential difference between point a and point b in volt is	(4)	zero
		(5)	2

Codes:

(A) P–1, Q–2, R–3, S–4 (B) P–2, Q–1, R–4, S–5

(C) P–1, Q–3, R–2, S–3 (D) P–1, Q–2, R–4, S–3

Space for Rough work

9. List-I contains units/dimensions and List-II some combination of quantities. Match the List-I to List-II.

List–I		List–II	
(P)	Henry	(1)	$\sqrt{LC}$ where L is inductance and C is capacitance
(Q)	Second	(2)	$\frac{M\ell^2}{Q^2}$, ℓ is length, M-mass and Q is charge
(R)	$(Second)^3$	(3)	RC, R is resistance & C is capacitance.
(S)	Dimensionless	(4)	C^2LR, C-capacitance, L = Inductance, R is resistance
		(5)	$\frac{Pc}{Q}$ where Q is radiation energy striking unit area pr second. c is speed of light and P is radiation pressure.

Codes:

(A) P–2; Q–1,3; R–4; S–5 (B) P–5; Q–1, 3; R–2; S–4

(C) P–5; Q–3; R–1, 4; S–2 (D) P–2; Q–1; R–4; S–3, 4

10. Velocity of sound in air is V and the length of organ pipe is L. P_o is the mean pressure and ΔP_0 is amplitude of pressure variations.

List – I		List – II	
(P)	If the pipe is closed at one end, the frequency of first overtone is	(1)	3V/4L
(Q)	If the pipe is open at both ends the frequency of fourth harmonic is	(2)	2V/L
(R)	If the pipe is closed at one end the pressure at open end is	(3)	0
(S)	The pressure at closed end is	(4)	$[P_o - \Delta P_o, \quad P_o + \Delta P_o]$
		(5)	P_o

Codes:

(A) P–1, Q–2, R–4, S–5 (B) P–1, Q–2, R–5, S–4

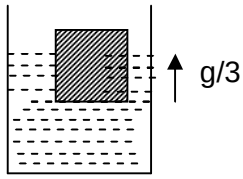
(C) P–2, Q–1, R–5, S–4 (D) P–3, Q–2, R–5, S–4

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. A cubical block is floating in a liquid with half of its volume immersed in the liquid. When the whole system accelerates upwards with a net acceleration of g/3. Find the fraction of volume immersed in the liquid

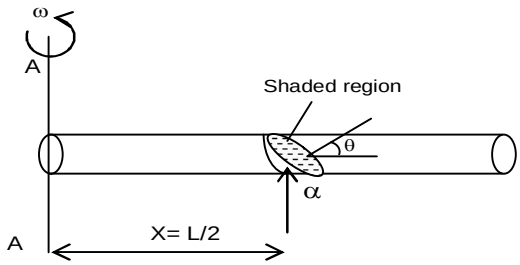


12. A projectile is thrown with a velocity of 18 m/s at an angle of 60° with horizontal. The interval between the moments when speed of the projectile is 15 m/s is ($g = 10\ m/s^2$)

13. A source of sound emitting a note of frequency 200Hz moves towards an observer with a velocity v equal to the velocity of sound. If the observer also moves away from the source with the same velocity v, the apparent frequency heard by the observer is

14. If 10% of the current passes through a moving coil galvanometer of resistance 99 ohm, then the shunt resistance will be:

15. A thin uniform metallic rod of mass m, length L, Young modulus of elasticity Y and cross–sectional area A is rotated by angular velocity ω about extreme end AA'. Consider a section on the rod at midpoint of rod. What will be the normal stress in N/m^2 on the shaded region? (Take the value of $\frac{M\omega^2L^2\cos^2\theta}{A} = 10\ N/m^2$)

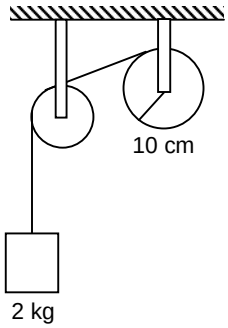


Space for Rough work

16. An object is cooled from 75^0C to 65^0C in 2 minutes in a room at 30^0C. The time taken to cool another identical object from 55^0C to 45^0C in the same room in minutes is

17. The thickness of a glass plate is measured to be 2.17 mm, 2.17 mm and 2.18 mm at three different places. Find the average thickness of the plate from this data. (in mm)

18. A string is wrapped on a wheel of moment of inertia 0.20 kg m^2 and radius 10 cm and goes through a light pulley to support a block of mass 2.0 kg as shown in figure. Find the acceleration of the block. (in m/s^2)



Space for Rough work

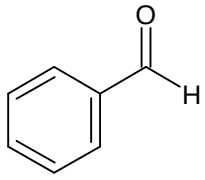
Chemistry

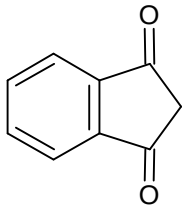
PART – II

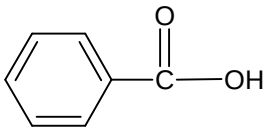
SECTION – A
(One or More than one correct type)

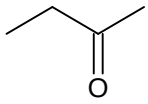
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Which of the following compounds exhibit acid base reaction with NaOH?

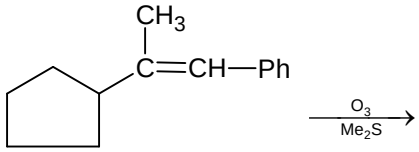
(A) 

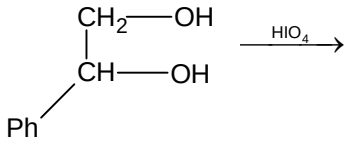
(B) 

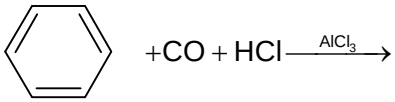
(C) 

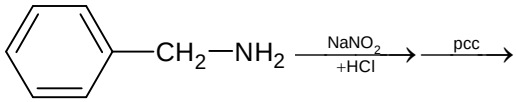
(D) 

20. Which of the following reactions can produce benzaldehyde as major product?

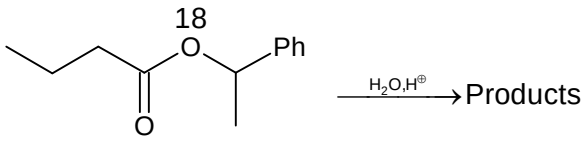
(A) 

(B) 

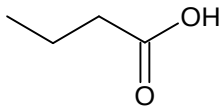
(C) 

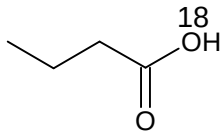
(D) 

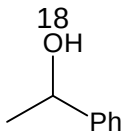
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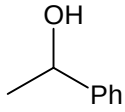
21. 

Products are:

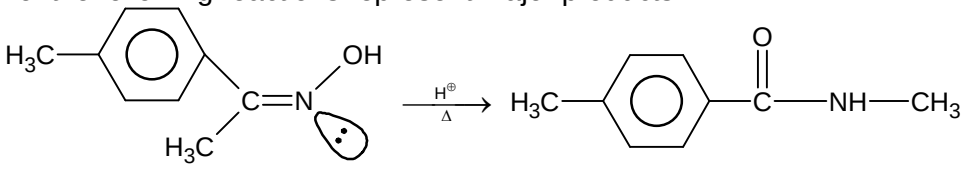
(A) 

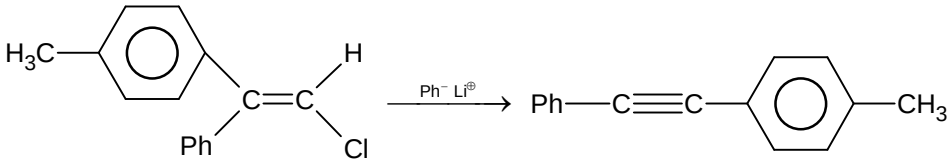
(B) 

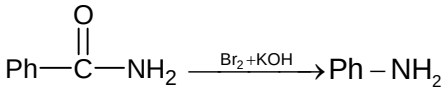
(C) 

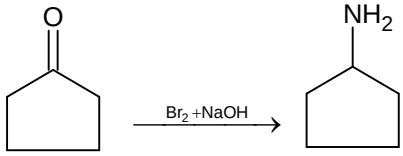
(D) 

22. Which of the following reactions represent major products?

(A) 

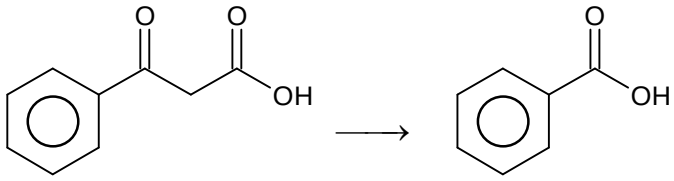
(B) 

(C) 

(D) 

Space for Rough work

23. The following conversion reaction can be carried out by using reaction sequences:



(A) $\xrightarrow{Zn-Hg/HCl} \xrightarrow{Br_2,hv} \xrightarrow{KCN} \xrightarrow{H_3^{\oplus}O,\Delta}$ (B) $\xrightarrow{NaBH_4} \xrightarrow{Al_2O_3,\Delta} \xrightarrow{O_3/H_2O(Oxidation)}$

(C) $\xrightarrow{\Delta} \xrightarrow{I_2+NaOH} \xrightarrow{H^{\oplus}}$ (D) $KMnO_4 / \bar{O}H/ \Delta$

24. Which of the following statements are correct?
(A) Hydrolysis of sucrose with dilute acid yields an equimolar mixture of D-Glucose and D-Fructose.
(B) Acidic hydrolysis of sucrose is accompanied by a change in optical reaction.
(C) In sucrose, the glycosidic linkage is between C – 1 of glucose and C – 2 of fructose.
(D) Aqueous solution of sucrose exhibits mutarotation.

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST–I** and **LIST–II**. **FOUR** options are given representing matching of elements from **LIST–I** and **LIST–II**. **ONLY ONE** of these four options corresponds to a correct matching.

25. Match List-I with List-II and select correct code for your answer.

List – I (Molecules)		List – II (M = Central atom)	
(P)	H_2O	(1)	largest M – H length
(Q)	H_2S	(2)	largest $H\hat{M}H$ bond angle
(R)	H_2Se	(3)	lowest boiling point
(S)	H_2Te	(4)	4p – 1s overlapping

Code:

	P	Q	R	S
(A)	3	2	4	1
(B)	2	3	4	1
(C)	3	1	4	2
(D)	1	3	4	2

Space for Rough work

26. List-I contains some cations and List-II contains some anion which produce white ppt with corresponding cation. Match List-I with List-II and select correct code for your answer.

List – I		List – II	
(P)	Ag^{+}	(1)	Cl^{-}
(Q)	Zn^{2+}	(2)	S^{2-}
(R)	Mg^{2+}	(3)	CO_3^{2-}
(S)	Ba^{2+}	(4)	SO_4^{2-}

Code:

	P	Q	R	S
(A)	2	3	1	4
(B)	4	3	2	1
(C)	1	4	3	2
(D)	1	2	3	4

27. The unbalanced chemical reactions given in List-I show missing reagent or condition (?) which are provided in List-II. Match List-I with List-II and select the correct answer using the code given below the lists:

List – I		List – II	
(P)	$PbO_2 + H_2SO_4 \xrightarrow{?} PbSO_4 + O_2 + \text{other product}$	(1)	NO
(Q)	$Na_2S_2O_3 + H_2O \xrightarrow{?} NaHSO_4 + \text{other product}$	(2)	I_2
(R)	$N_2H_4 \xrightarrow{?} N_2 + \text{other product}$	(3)	Warm
(S)	$XeF_2 \xrightarrow{?} Xe + \text{other product}$	(4)	Cl_2

Code:

	P	Q	R	S
(A)	4	2	3	1
(B)	3	2	1	4
(C)	1	4	2	3
(D)	3	4	2	1

Space for Rough work

28. Match List-I with List-II.

List – I (Metal)		List – II (Outer electronic configuration of either metal or its M^{2+} ion)	
(P)	Cu	(1)	ns^2, np^6, nd^{10}
(Q)	Cd	(2)	ns^2, np^6, nd^9
(R)	Fe	(3)	$(n-1)d^5, ns^2$
(S)	Mn	(4)	ns^2, np^6, nd^6

Code:

	P	Q	R	S
(A)	1	2	3	4
(B)	2	1	4	3
(C)	2	1	3	4
(D)	1	2	4	3

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. For soaps critical micelle concentration (CMC) is 10^{-x} (min.) to 10^{-y} (max.) mol/L. What is the value of x?

30. The average life of a radioactive element is 7.2 min. Calculate the time required (in min.) for the stage of 33.33% decay ($\log 3/2 = 0.1761$)

31. 0.16 g N_2H_4($K_b = 4 \times 10^{-6}$) are dissolved in water and the total volume made up to 500 mL. Calculate the mole fraction of N_2H_4 that has reacted with water in this solution.

Space for Rough work

32. At 273 K and 1 atm, 10 litre of N_2O_4 decomposes to NO_2 according to equation

$N_2O_2(g) \rightleftharpoons 2NO_2(g)$

What is degree of dissociation (α) when the original volume is 25% less than that of existing volume?

33. What will be the pH of an aqueous solution of 1.0 M ammonium formate?
Given : $pK_a = 3.8$ and $pK_b = 4.8$

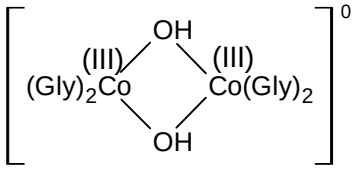
34. For the reaction; $FeCO_3(s) \longrightarrow FeO(s) + CO_2(g); \Delta H = 82.8$ kJ at 25°C, what is (ΔE or ΔU) at 25°C?

35. Calculate $|\Delta_f H°|$ (in kJ/mol) for Cr_2O_3 from the $\Delta_r G°$ and the S° values provided at 27°C

$4Cr(s) + 3O_2(s) \longrightarrow 2Cr_2O_3(s); \quad \Delta_r G° = -2093.4$ kJ / mol

$S°(J/K\ mol): S°(Cr,s) = 24; \ S°(O_2,g) = 205; \quad S°(Cr_2O_3,s) = 81$

36. Total number of Co – N (cobalt-nitrogen) linkages in



(Gly^{-1} = Glycinate)

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If $z_1 = a + ib$ and $z_2 = c + id$ are two complex numbers such that $|z_1| = |z_2| = 1$ and $\text{Re}\left(z_1\bar{z}_2\right) = 0$, then the pair of complex numbers, $w_1 = a + ic$ and $w_2 = b + id$ satisfying

(A) $|w_1| = 1$ (B) $|w_2| = 1$

(C) $\left|w_1\bar{w}_1\right| = 1$ (D) $\text{Re}\left(\bar{w}_1 w_2\right) = 0$

38. If p, q, r, s are positive real numbers and the equation $x^4 - px^3 + qx^2 - rx + s = 0$ has four positive real roots, then

(A) $p^4 \geq 256s$ (B) $r^4 \geq 256s^3$

(C) $pr \geq 16s$ (D) $q^2 \geq 36s$

39. In a ΔABC with fixed base BC, the vertex A moves such than $\cos B + \cos C = 4\sin^2\left(\frac{A}{2}\right)$.

If a, b and c denote the length of the sides of the triangle opposite to the angle A, B and C respectively, then

(A) $b + c = 4a$

(B) $b + c = 2a$

(C) locus of point A is an ellipse

(D) locus of point A is a pair of straight lines.

40. Let $f(x) = \begin{cases} x^3 + x^2 - 10x & -1 \leq x < 0 \\ \sin x & 0 \leq x < \frac{\pi}{2} \\ 1 + \cos x & \frac{\pi}{2} \leq x \leq \pi \end{cases}$ then $f(x)$ has

(A) local maxima at $x = \frac{\pi}{2}$ (B) local minima at $x = \frac{\pi}{2}$

(C) absolute maxima at $x = 0$ (D) absolute maxima at $x = \frac{\pi}{2}$

Space for Rough work

41. If $I_n = \int_{-\pi}^{\pi} \frac{\sin nx}{(1+\pi^x)\sin x} dx, = 0,1,2$ then

(A) $I_n = I_{n+2}$ (B) $\sum_{m=1}^{10} I_{2m+1} = 10\pi$

(C) $\sum_{m=1}^{10} I_{2m} = 0$ (D) $I_n = I_{n+1}$

42. If $f(x) = \begin{cases} \int_0^x (1+|1-t|)dt, x > 2 \\ 5x+1, x \le 2 \end{cases}$, then

(A) f(x) is not continuous at x = 2
(B) f (x) is continuous but not differentiable at x = 2
(C) f (x) is differentiable everywhere
(D) the right derivative of $f(x)$ at x = 2 does not exist

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST–I** and **LIST–II**. **FOUR** options are given representing matching of elements from **LIST–I** and **LIST–II**. **ONLY ONE** of these four options corresponds to a correct matching.

43. Match the sequence $a_1, a_2, a_3, ...$ whose nth term is given on the left with properties of the sequence on the right

LIST - I		LIST – II	
(P)	$a_n = \int_0^{\pi/2} \frac{1-\cos 2nx}{1-\cos 2x} dx$	(1)	a_1, a_2, a_3 … are in A.P.
(Q)	$I_n = \int_0^{\pi/4} \tan^n x\, dx \quad a_n = \frac{1}{I_{n+1} + I_{n+3}}$	(2)	$a_1, a_2, a_3, ...$ are in G.P.
(R)	$I_n = \int_0^{\pi/4} \frac{\sin^2 nx}{\sin x} dx \forall n \ge 0$ and $a_n = I_n - I_{n-1} \forall n \ge 1$	(3)	$a_1, a_2, a_3, ...$ are in H.P.
(S)	$b_n = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x} dx \quad a_n = b_n - b_{n-1} \forall n \ge 1$	(4)	$a_1, a_2, a_3, ...$ is a constant sequence.
		(5)	$a_1, a_2, a_3, ...$ are in A.G.P.
		(6)	$a_1, a_2, a_3, ...$ are neither in A.P. nor G. P nor H.P.

The correct option is

(A) P → 1; Q → 1; R → 3; S → 1,2, 3, 4
(B) P → 5; Q → 2; R → 6; S → 1
(C) P → 1; Q → 1; R → 3; S → 6
(D) P → 1; Q → 1; R → 5; S → 1, 2, 3, 4

Space for Rough work

44. Let A be a set containing n elements. P and Q are two subsets of A. The number of ways of choosing P and Q so that

LIST - I		LIST – II	
(P)	$P \cup Q = A$	(1)	2^{n-1}
(Q)	$P \cup Q \neq \phi$	(2)	3^n
(R)	$P \sim Q$ is a singleton	(3)	$2^n + 1$
(S)	$P \cap Q = \phi$	(4)	$3^n - 1$
		(5)	$n.3^{n-1}$
		(6)	$4^n - 3^n$

The correct option is
(A) P → 2; Q → 6; R → 1; S → 5
(B) P → 1; Q → 2; R → 3; S → 4
(C) P → 2; Q → 6; R → 5; S → 2
(D) P → 1; Q → 2; R → 5; S → 6

45. Match the statement given in List I with List II

LIST – I		LIST – II	
(P)	The set $\left\{Re\left(\frac{2iz}{1-z^2}\right) : z \text{ is a complex number}, \lvert z\rvert = 1, z \neq \pm 1\right\}$	(1)	$(-\infty, -1) \cup (1, \infty)$
(Q)	The domain of the function $f(x) = \sin^{-1}\left(\frac{8(3)^{x-2}}{1-3^{2(x-1)}}\right)$ is	(2)	$(-\infty, 0) \cup (0, \infty)$
(R)	If $f(\theta) = \begin{vmatrix} 1 & \tan\theta & 1 \\ -\tan\theta & 1 & \tan\theta \\ -1 & -\tan\theta & 1 \end{vmatrix}$, then the set $\left\{f(\theta) : 0 \leq \theta < \frac{\pi}{2}\right\}$ is	(3)	$[2, \infty)$
(S)	If $f(x) = x^{3/2}(3x-10)$, $x \geq 0$, then $f(x)$ is increasing in	(4)	$(-\infty, -1] \cup [1, \infty)$
		(5)	$(-\infty, 0] \cup [2, \infty)$
		(6)	$(-\infty, -2] \cup [-1, \infty)$

The correct option is
(A) P → 4; Q → 5; R → 3; S → 3
(B) P → 3; Q → 3; R → 4; S → 5
(C) P → 4; Q → 3; R → 3; S → 5
(D) P → 4; Q → 3; R → 2; S → 1

Space for Rough work

46. Suppose a. b, c are three distinct real numbers and p (x) is a real quadratic polynomial such that $\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix}\begin{bmatrix} p(-1) \\ p(1) \\ p(1) \end{bmatrix} = \begin{bmatrix} 3a^2+3a \\ 3b^2+3b \\ 3c^2+3c \end{bmatrix}$

LIST - I		LIST – II	
(P)	x – coordinates (s) of the point of intersection of $y = f(x)$ with the x – axis	(1)	4
(Q)	Area bounded by $y = \frac{3}{2} f(x)$ and the x – axis	(2)	2
(R)	Maximum value of $f(x)$	(3)	1
(S)	Length of the intercept made by $y = f(x)$ on the x – axis	(4)	–2
		(5)	0
		(6)	3

The correct option is
(A) P → 2, 4; Q → 1; R → 3; S → 1
(B) P → 1, 2; Q → 2; R → 3; S → 4
(C) P → 5; Q → 6; R → 3; S → 1
(D) P → 5, 6; Q → 1; R → 3; S → 1

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. If $x_n = 1^2 + (2)(2^2) + 3^2 + (2)(4^2) + ... = n\frac{(n+1)^2}{2}$ if n is even, find x_{51}

48. Find the number of three digit numbers of the form xyz with x, z < y, $x \neq 0$.

Space for Rough work

49. Let $a_r = r^2 \frac{^{50}C_r}{^{50}C_{r-1}} a$ and b denote the coefficient of x^{49} in $(x-a_1)(x-a_2)..(x-a_{50})$, find $\left(-\frac{1}{17}\right)b$.

50. Let ABC and ABC' be tow non – congruent triangles with sides AB = 4, AC = AC' = $2\sqrt{2}$ and angle B = 30°. The absolute value of the difference between the areas of these triangles is

51. In a triangle ABC; A = $\frac{\pi}{3}$, b = 50, c = 30, AD is a median through A, then AD^2 is equal to

52. The tangent at three points A, B, C on the parabola $y^2 = 4x$, taken in pairs intersect at the point P, Q and R. If Δ, Δ' be the areas of the triangles ABC and PQR respectively, then $\frac{\Delta}{\Delta'} =$

53. If the position vector of the point of intersection of the line $r = (i + 2j + 3k) + \lambda(2i + j + 2k)$ and the plane $r.(2i - 6j + 3k) + 5 = 0$ is $a\hat{i} + b\hat{j} + c\hat{k}$ then value of $(50a + 60b + 75c)^2$ is equal to

54. If $f(x) = \frac{\sin^{-1}(1-\{x\}).\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}.(1-\{x\})}$, then the value of $\left(\frac{18\sqrt{2}}{\pi} \lim_{x\to 0^-} f(x)\right)$ is (where {x} denotes the fractional part of x)

Space for Rough work